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# The Composite Overfit Analysis Framework: Assessing the Out-of-Sample Generalizability of Construct-Based Models Using Predictive Deviance, Deviance Trees, and Unstable Paths

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**Abstract.** Construct-based models have become a mainstay of management and information systems research. However, these models are likely overfit to the data samples upon which they are estimated, making them risky to use in explanatory, prescriptive, or predictive ways outside a given sample. Empirical researchers currently lack tools to analyze why and how their models may not generalize out of sample. We propose a composite overfit analysis (COA) framework that applies predictive tools to describe the sources and ramifications of overfit in terms of the focal concepts important to empirical researchers: cases, constructs, and causal paths. The COA framework begins by using a leave-one-out crossvalidation procedure to identify cases with unusually high predictive error given their in-sample fit—a difference we describe as *predictive deviance*. The framework then employs a novel *deviance tree* method to group deviant cases that have similar predictive deviance and for similar theoretical reasons. We then employ a leave-deviant-group-out method, which sequentially analyzes how each *deviant group* affects model parameters, thereby identifying potentially *unstable paths* in the model. We can then infer descriptive reasons for why and how overfit affects a given model and data sample using the grouping criteria of the deviance tree, construct scores of deviant groups, and resulting unstable paths. These insights allow researchers to identify unexpected behavior that could define boundary conditions of their theory or point to new theoretical phenomena. We demonstrate the practical utility of our analytical framework on a technology adoption model in a new context.

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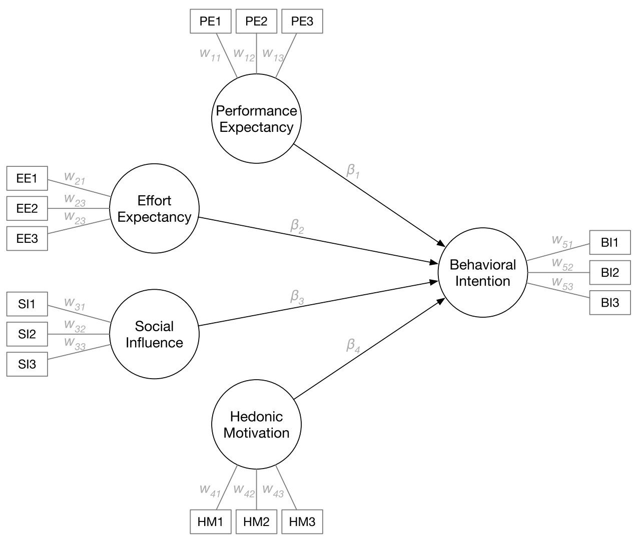
## 1. Introduction

Researchers often use construct-based modeling to analyze structures of relationships between conceptual variables that are measured by multiple measurement items (Gefen et al. 2011). Models using these multi-item variables, known as constructs, are typically estimated using covariance-based structural equation modeling (CB-SEM; such as LISREL™) (Jöreskog and Sörbom 1996) or composite-modeling techniques, such as partial least squares path modeling (PLS-PM) (Wold 1975). Construct-based modeling once constituted nearly a fifth of the studies published in top-tier management information systems journals (Urbach and Ahlemann 2010) and remains a popular method to this day (Hair et al. 2017a). Figure 1 shows a simple example of a construct-based model with five constructs (circles) that are each measured by three measurement items (rectangles) on a survey.

The five constructs in this simple model are drawn from the larger and well-regarded UTAUT2 model (Venkatesh et al. 2012) that seeks to explain why users might adopt and use an information technology based on prior expectations and perceptions. Our simplified model employs four antecedent explanations of behavioral intentions. Constructs represent abstract concepts, such as performance expectancy, and are measured by multiple manifest measurement items (such as survey questions) that can either reflect or define the construct. Each construct is related to its measurement items by either a weight coefficient $w_{ij}$, as in our example, or a loading $\lambda_{ij}$. Four of the constructs have structural

**Figure 1.** Simple Example of a Construct-Based Model




relationships with a single outcome construct, estimated as path coefficients $\beta_i$. The use of constructs provides a parsimonious structural model with hypothesized relationships between a small number of constructs rather than a large number of items.

Researchers evaluate construct-based models by testing their fit to a data sample and examining the magnitude and significance of path coefficients, from which they infer the validity of hypothesized relationships between constructs. However, researchers rarely consider how well their construct-based models can predict outcomes for new observations. For example, a marketing researcher might apply our simple model in Figure 1 to explain how behavioral intention to use an information system arises from users' expectancy and perceptions and then, estimate the structural paths using sample data from hundreds of current customers. Would these estimated relationships hold for a future or different sample of customers? Current modeling practices favor extracting a story from the data sample on hand without considering how it might change for new or future data. This neglect of out-of-sample generalizability hampers the generalizability and utility of theory: "It remains true that if we can predict successfully on the basis of a certain explanation, we have good reason and perhaps the best sort of reason, for accepting the explanation" (Kaplan 1964, p. 350). This concern is particularly salient for construct-based models because they are increasingly believed to be overfit to the sample of observations, or *cases*, on which they are estimated (McIntosh et al. 2014).

Overfit implies that predictions generated from estimated models could yield surprisingly large errors if used to prescribe actions for new or future cases. Such practical ramifications might not seem at first to relate to explanatory studies at the average or aggregate level. However, we will demonstrate that the performance gap between in-sample fit and out-of-sample prediction also exposes limitations of the theories used to develop such models and can render some explanations invalid altogether. At present, however, the construct-based modeling literature lacks methods to diagnose overfit problems in construct-based models and derive what such problems imply about theory. Researchers need new tools to quantitatively identify the sources of overfit and translate their implications into descriptive elements that are relevant to theory and practice: cases, constructs, and theorized relationships.

We have seen nascent efforts to address the out-of-sample generalizability of construct-based models in the form of methods to generate predictions and assess the predictive relevance of such models (Evermann and Tate 2016; Shmueli et al. 2016, 2019; Danks and Ray 2018; Danks 2021). However, these purely predictive methods face a series of limitations that hamper the utility of their predictions, validations, and implications. At an operational level, the predictive methods in recent literature only apply to item-level measurements, whereas empirical researchers are primarily interested in the relationships between constructs. Moreover, trying to validate the predictive utility of construct-based models runs counter to the logic and practices of modern predictive methodologies, which seek to tweak parameters to maximize prediction accuracy rather than make categorical statements about predictive power. Thus far, it has not been clear what role prediction might play in construct-based studies or how it can advance theory (Rönkkö et al. 2016).

In this study, we build an analytic framework that applies predictive tools to examine the out-of-sample generalizability of construct-based models through the lens of overfit—the composite overfit analysis (COA) framework. The COA framework describes the sources and ramifications of overfit at the focal level of cases, constructs, and causal paths familiar to empirical researchers. When applied to a model and corresponding sample of data, the COA framework first identifies groups of cases that have in-sample and out-of-sample errors that are considerably different. Each deviant group can be quantitatively described and discriminated from other groups by identifying common patterns and differences in their construct scores that run counter to theory. We can also investigate how these groups affect structural paths and lead to overfit, thus pointing to hypothesized interconstruct relationships that might suffer poor out-of-sample generalizability.

In doing this, the COA framework aligns with many of the emerging criticisms of inferential practices and urgent calls for reform: moving beyond categorical notions of fit and significance (Amrhein et al. 2019, Hair et al. 2019), understanding when and why theory fails (Gray and Cooper 2010), and introducing more predictive methods (Shmueli and Koppius 2011). A common thread in these calls for change is for researchers to better understand the nuances of the cases in their data and to address new forms of uncertainty. In this spirit, we aim to give researchers tools to tell more nuanced stories, to draw attention to groups of cases of special interest, and to deal with predictive uncertainty in more straightforward ways. The COA framework does not replace inferential conclusions, nor is it a purely predictive exercise. Rather, it marries both approaches into an explanatory-predictive framework that capitalizes on the strengths of each approach.

After introducing the COA framework, we will demonstrate it on a technology adoption and use model applied to a data set of responses from users evaluating personal task management software. The empirical demonstration highlights unusual groups of cases and potentially untheorized phenomena that have theoretical and practical implications. The format of the demonstration might serve as a template for applying the COA framework to other models and data.

# 2. Predictive Methodologies and Construct-Based Research

## 2.1. Winds of Change

Our study is motivated first by concerns that construct-based models are overfit and may suffer poor generalizability when applied to out-of-sample cases. Although inferential generalizability relates to how close estimated parameters are to population parameters, predictive generalizability relates to the ability of the model to accurately predict outcomes for cases that were not used to estimate parameters. Predictive methods attend to the concerns of out-of-sample generalizability with an emphasis on overfit and case-level analysis.

At the same time, we also paid heed to emerging concerns around empirical studies and specifically, construct-based modeling. Whereas construct-based modeling depends heavily on categorical tests of model fit and coefficient significance, recent years have seen a firestorm of debate about the future of such inferential studies and calls for methodological reform grow louder. At the broadest level of scientific thought, scientists are calling for inferential research to move away from categorical validations, such as significant versus nonsignificant, to telling more nuanced stories about different types of uncertainty (Amrhein et al. 2019). In the management realm, researchers are further advocating for new research methods that examine cases that do not fit theory and from them, describe the conditions under which and the reasons why a proffered theory fails (Gray and Cooper 2010). One lucrative route to addressing both these calls is to temper pure explanatory modeling with predictive approaches (Shmueli and Koppius 2011).

Applying predictive methods provides an opportunity to address many of these concerns. Unlike extant inferential practices, predictive methods and practices avoid binary tests for good or bad performance and categorical validation of models. Prediction describes model utility in terms of generalizability to new and unseen data (Gregor 2006) and relevance to practical decision making (Shmueli et al. 2019). Predictive methods also care about case-level analysis (Shmueli et al. 2016), allowing more fine-grained analysis of smaller sets of cases than inferential methods that describe the average case. For these reasons, we believe that extending predictive methods allows us to address many of these concerns while developing a framework for assessing and addressing out-of-sample generalizability issues.

## 2.2. Predicting with Constructs

A major limitation of introducing predictive methods to construct-based modeling is that there are conceptual misalignments between the two approaches that have forestalled the promise of using them together. We seek to realign the conceptual goals of predictive analysis with those of construct-based theory development. Mixing predictive methods with the inferential approaches of construct-based studies is fraught with conflation of terminology and a mismatch in objectives. To start, we distinguish and clarify the goals and terminologies of construct-based inferential and predictive methodologies.

**Constructs, Not Items.** Given the game-changing potential of predictive methods, it is perhaps unsurprising that there has been a recent spate of methodological studies seeking to create predictive solutions for construct-based models (Evermann and Tate 2016; Shmueli et al. 2016, 2019; Danks and Ray 2018; Danks 2021). Constructs, such as those in our simple example in Figure 1, are often abstract concepts that are not directly observable and must be measured by one or more observable measurement items. Consequently, recent predictive studies have largely sought to evaluate the predictive performance of the *measurement items* of outcome constructs, as predicted from the measurement items of antecedent constructs. Focusing on measurement items is in line with traditional predictive practices because validating predictions requires comparing predicted outcomes against observable, actual outcomes. However, the theoretical interests of researchers building construct-based models lie in their domain-relevant constructs. Thus, the recent predictive methods focusing on measurement items do not help theoreticians to draw insight on what they care about most: constructs and the relationships between constructs. Given the primacy of constructs in theory building, one major contribution of this study is to devise ways of assessing the prediction of theorized constructs rather than measurement items.

**Composites, Not Factors.** Applying predictive methods to constructs requires careful examination of how constructs are conceived and operationalized. Empirical researchers model constructs as either common factors or as composites (Rigdon 2012, Henseler et al. 2014, Sarstedt et al. 2016, Hair et al. 2017b). Causal relationships in common factor models are estimated using covariance-based techniques that conceive constructs as being latent concepts without determinable scores. However, if we are to apply predictive methods to constructs, we require construct scores to be measurable for both in-sample and out-of-sample data. We are not aware of any means of obtaining determinable scores for common factors, and so, our analyses of overfit, out-of-sample generalizability, and the ensuing COA framework cannot be immediately applied to common factor methods, such as CB-SEM.

Composite scores, on the other hand, are simply weighted sums that use measurement weights estimated from sample data. The ready availability of determinable scores has made composites the measurement model of choice for recent prediction-oriented literature on construct-based models (Ringle et al. 2012; McIntosh et al. 2014; Henseler et al. 2016; Shmueli et al. 2016, 2019; Hair et al. 2019). Construct-based models using composite constructs also constitute more than half of the construct-based studies in top information systems journals (Urbach and Ahlemann 2010) and are estimated using methods such as PLS-PM (Rigdon 2012, Sarstedt et al. 2016).

We continue in the tradition of calling for prediction-oriented analysis to use composite measurement models. We also follow the common practice of using PLS-PM to estimate composite-based models, but we expect that many of the proposed techniques are adaptable to newly emerging forms of composite modeling methods, such as general structured component analysis (GSCA) (Hwang and Takane 2004, Hwang 2009) and regularized generalized canonical correlation analysis (RGCCA) (Tenenhaus and Tenenhaus 2011).

## 2.3. Reconciling Fit and Prediction

**Fitting Vs. Predicting.** The goals and practices of explanatory modeling differ from those of predictive modeling. There is a potential for confusion in the language of the two practices, and so, we need to find common ground if we are to link predictive and inferential methods. We, therefore, devote effort to distinguishing the key terms of the two practices before integrating their goals.

Theory-driven research using construct-based modeling has traditionally emphasized parameter estimation and *model fit* as key components of empirical validation of theory (Hair et al. 2019). A theoretically-justified statistical model is fit to a sample of cases by estimating the model's parameters from that sample. Just as in regression, the estimated parameters of composite models can be used to calculate fitted scores for outcomes. Methodological texts often refer to antecedents as predictors and fitted outcome scores as predictions (Wold 1982, Chin 1998, Rigdon 2012). However, the term prediction has more recently developed its own nuance in predictive methodologies as an out-of-sample estimated score. Thus, for the sake of clarity, we will refer to fitted outcome scores that are calculated *in sample* (i.e., from the same data the model's parameters were estimated from) as *fitted scores*. We will use the notation $\hat{Y}_{(in)}$ for fitted scores, where *in* denotes that it is calculated on in-sample data, akin to training data from a predictive perspective.

The difference between the actual outcome value and the fitted score is the *fit error*. Model fit is often operationalized as metrics that aggregate fit error across all cases. In this paper, we measure model fit as the mean square of fit errors or $MSE_{(in)}$. The popularly used $R^2$ metric, which evaluates variance explained by the model, incorporates $MSE_{(in)}$ as the variance unexplained (or error variance).

In contrast, predictive modeling is generally concerned with how well a model estimates outcome scores of new cases rather than the training cases used to estimate the model. In this paper, we will strictly use the term *prediction* to describe when a model trained to fit one set of cases (in-sample data) is used to generate the outcome scores of *out-of-sample* or *validation* cases that were not used in training. We will use $\hat{Y}_{(out)}$ for predicted scores where *out* denotes that it is calculated on out-of-sample data. When a predictive model is being validated, a typical approach is to partition cases into training (in-sample) and validation (out-of-sample) data sets (Arlot and Celise 2010).

The difference between the out-of-sample actual outcome and the predicted outcome score is the *prediction error*. Predictive metrics aggregate prediction error across all the cases of the out-of-sample data set only. In this paper, we will use $MSE_{(out)}$ as an aggregate of prediction error that is used to evaluate *prediction accuracy*. Metrics, such as $MSE_{(out)}$, indicate how well a fitted model can predict new data. Note that prediction error is generally expected to be larger than fit error because prediction error is affected by both model misspecification (error bias) and the added uncertainty of predicting new cases (error variance) (Hastie et al. 2013). Thus, $MSE_{(out)}$ is a better indicator than $MSE_{(in)}$ of how well a model generalizes to new data.

**Overfit.** The contrast between fit error and prediction error reveals a connection between the practices of inferential modeling and predictive modeling. As models become too closely fitted to one sample, they often become worse at predicting outcomes of other samples (Hastie et al. 2013). Thus, enhancing a model's fit to an in-sample data set does not always yield better prediction;

it can lead to a model that is *overfit* to these cases. Overfit is where the concepts of fit and prediction meet and intertwine. The natural tension and trade-off between fit and prediction become an interesting boundary where we can explore how prediction can inform theory.

We expect that understanding and evaluating the ramifications of overfit can offer insights to theory-driven research that is both concerned with explaining phenomena by fitting empirical models as well as about the practical implications of how that model generalizes to new or future cases. We can evaluate the extent of model overfit by comparing fit error and prediction error to quantify the loss in accuracy when generalizing out of sample (Hastie et al. 2013).

There has been little to no emphasis on evaluation of overfit in construct-based models in the literature, despite recent advancements of predictive methods in the field (Shmueli et al. 2019). This neglect is particularly worrying given that studies applying composite techniques, such as PLS-PM, routinely rely on in-sample average-case criteria, such as $R^2$. These criteria routinely favor overly complex models that overfit the data by tapping into idiosyncratic patterns of a specific sample (Sharma et al. 2019). Although all construct-based models are at risk for overfit, the topic of overfit is particularly salient in composite modeling techniques like PLS-PM, which aggregate items laden with measurement error and incorporate this error into the structural regression. As such, composite-based models may be particularly susceptible to overfitting (McIntosh et al. 2014, p. 228).

Beyond measurement error, overfit in composite-based models is also tied to their high complexity in terms of parameterization, small sample size, and common practices of maximizing model fit by modifying and adjusting model parameters. James et al. (2013, p. 144) note that "[t]he higher the ratio of parameters $p$ to number of samples $n$, the more we expect overfitting to play a role."

Complexity in construct-based models arises from both structural and measurement model parameters, which need to be estimated. The structural parameters specify hypothesized relationships between constructs of interest, including control variables. At their simplest, structural models are comparable with the regression models found in the management literature. However, structural models might also contain features such as mediated relationships that are not found in a single regression model. The measurement portion of a construct-based model adds a large number of parameters to the overall model, with constructs being measured by multiple items that each require an additional parameter relating the item to its construct. Moderated relationships, modeled using a new construct to represent the interaction between two constructs, have a multiplicative effect on the number of parameters because they require every combination of items across the two constructs to be a new item-pair product whose measurement weights require estimation.

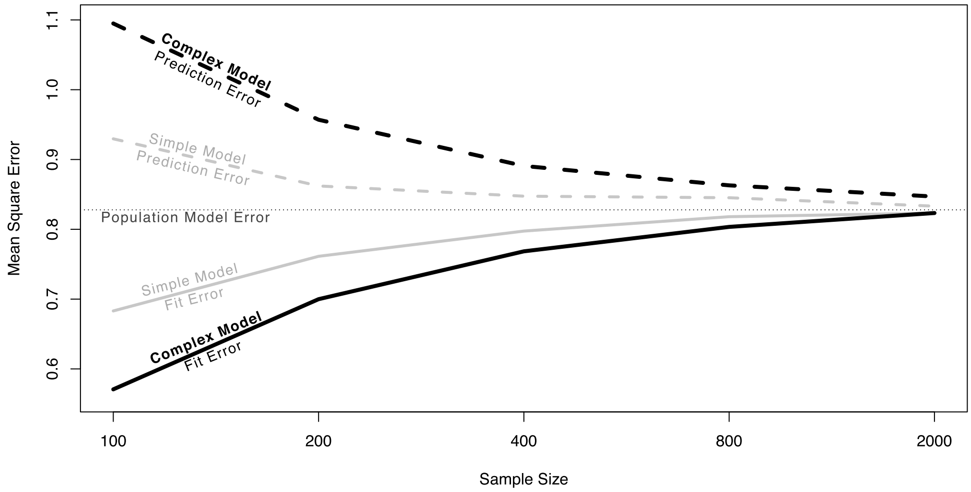
Thus, a typical construct-based model for information systems research (based on criteria by Hair et al. 2017a), such as the technology acceptance model 2 (Venkatesh and Davis 2000) with seven antecedent constructs, three mediating constructs, three interaction terms, and an outcome construct, has over 55 parameters to be estimated. Unlike common factor models, composite models also do not constrain associations between items of a construct (McIntosh et al. 2014). These factors all contribute to models that are considerably more complex than those in typical regression studies, despite often involving a similar number of theorized concepts and hypothesized relationships.

To make matters worse for overfit, construct-based models are often estimated using modestly sized samples, with an average of around 300 cases in fields like information systems (Hair et al. 2017a). Overfit is directly related to sample size relative to model complexity; as the number of parameters approaches the size of the data set, the model becomes increasingly overfit (Hastie et al. 2013). Consequently, it comes as no surprise that composite models are believed to overfit their data.

To empirically observe the potential for PLS-PM to overfit to a data sample at hand, we conducted a simulation comparing a simple construct-based model with a complex one at various sample sizes using the R package cbsem (Schlittgen 2019, Schlittgen et al. 2020). The simple simulated model contains four exogenous constructs and one endogenous construct, just like our simple example model in Figure 1. The complex simulated model mimics the complexity seen in the literature (Hair et al. 2017a) and closer to the full UTAUT2 model, with 10 exogenous constructs and 1 endogenous construct, requiring 44 measurement parameters and 10 structural parameters. All constructs in the simple and complex models have four measurement items. We varied the reliability of our measures and considered sample sizes of 100, 200, 400, 800, and 2,000, like other simulation studies of composite models (Sharma et al. 2019). We repeated the simulation 1,000 times for each condition. For each simulation, we calculated in-sample fit error as $MSE_{(in)}$, and we generated predictions for the validation data to calculate an out-of-sample prediction error as $MSE_{(out)}$ and then, compared both with the known population mean squared error (MSE; unexplained variance) that is free of any specification error. The results of this simulation are summarized and illustrated in Figure 2, where population MSE and the $MSE_{(in)}$ and $MSE_{(out)}$ of simple and complex models are plotted as a function of sample size.

We observe that even the simple model (solid gray and dashed gray lines) consistently overfits for smaller data samples, as evidenced by the in-sample fit MSE considerably underestimating the true population error.

**Figure 2.** Overfit Plot Comparing Prediction Vs. Fit MSE of Simple and Complex Models




*Note.* Mean in-sample MSE (Fit Error), mean out-of-sample MSE (Prediction Error), and population model MSE (Population Model Error) across simulation conditions for simple and complex models are plotted as a function of sample size.

The overfit worsens for out-of-sample predictions, as evidenced by the out-of-sample prediction MSEs overestimating population error. That the in-sample and out-of-sample MSEs do not converge to the true population model error at a sample size of 800, even with a simple model, should surprise most researchers. We additionally observe the overfit penalty of having a complex model with a large number of parameters; its in-sample fit even further underestimates population error, and it has worse out-of-sample prediction error.

Recently, construct-based studies have justified their sample sizes using minimum sample size criteria suggested by methodological research. We considered these suggested minimum sample sizes for our simulated complex model using the methods offered in Faul et al. (2009), which suggests $n = 226$, and Kock and Hadaya (2018), which suggests $n = 613$. However, at both suggested sample sizes, the large gap between in-sample and out-of-sample MSEs suggests considerable overfit. Only at a sample size of 2,000 does the complex model no longer visibly exhibit more overfit than the simple model, and both models approach the true population error. Thus, obeying even the most robust minimum sample size calculations for statistical power does not adequately address concerns of overfit.

Many of the applied practices around construct-based modeling likely further exacerbate overfit beyond what our simulation suggests. For example, items are often screened using exploratory or confirmatory techniques and then removed based on results of preliminary validity tests conducted on the in-sample data. In some construct-modeling techniques, interitem covariances can also be freed for estimation, often based on fit-improvement criteria like modification indices. Such sample-specific approaches only improve a model's fit to the given sample and do not improve the model's fit to future data sets or other samples. These practices are at odds with predictive analytics methods, which are highly cognizant of overfit and use a separate validation data set for fine-tuning parameters (Hastie et al. 2013). We also note that the remedies typically applied in predictive methodologies to minimize overfit, such as employing different objective functions or optimization criteria and dropping or regularizing parameters, are contrary to the goals and aims of theoretical causal modeling, where theories determine construct measurement and structural relationships.

Overall, we posit that the complex nature of construct-based models and the practices surrounding them result in overfit models that likely do not generalize well out of sample. An analysis comparing a model's fit with its predictive performance can bring specific aspects of overfit to light.

**Understanding Overfit: Cases and Parameters.** Simply diagnosing the presence of overfit might not be valuable given that composite models are likely generally overfit. We expect far greater insight from understanding *why* overfit creeps into models and *how* it manifests itself. After all, construct-based models are primarily explanations of why and how phenomena manifest. To begin, we recognize that whereas model fitting seeks to model the average case, predictive methods are focused on the outcome of individual cases (Shmueli et al. 2016). Adopting a case-level perspective, we can think of overfit as out-of-sample misfit that increases when parameters of a model cater too strongly to particular cases of the in-sample data.

Investigating new theoretical and explanatory issues from a predictive lens requires new methods to identify cases and parameters involved in overfit. However, beyond identifying them, we must also be ready to conceptualize what characterizes these cases and interpret how their impact on model coefficients affects theory. In this manner, we hope that researchers can analyze the out-of-sample generalizability of their models with an eye to discovering theoretical limitations and opportunities.

## 3. The Composite Overfit Analysis Framework

The COA framework is a multistep approach to investigating the sources and ramifications of overfit in construct-based models and their associated data samples for the purpose of adding nuance to our theoretical explanation of phenomena. We begin by introducing metrics relating to the prediction accuracy and overfit of outcome constructs. We then apply novel predictive methods to uncover potential theoretical issues by (a) identifying groups of cases that exacerbate overfit, (b) characterizing groups of deviant cases, and (c) identifying structural relationships between constructs that may not generalize well out of sample. Put together, this information can help us understand the boundaries of a theory and how it can be extended.

The COA framework should be applied to a model that has already met the criteria for the validation of both the measurement model and the structural model and with a data set that has been subject to standard cleaning and quality checks (Hair et al. 2021). Although we do not discuss these steps in further detail here, it is important to ensure that neither heterogeneity, multicollinearity, nor poor measurement influence the framework's analysis.

To implement the COA framework, we must overcome certain methodological constraints and challenges. These challenges pertain to how to estimate prediction errors of constructs, how to identify individual or groups of cases in the data that perturb parameter estimates when they are left out of the sample, and how to identify structural relationships that might be overfit to the point of reducing out-of-sample generalizability.

### 3.1. Evaluating Prediction Accuracy and Overfit

**Prediction Accuracy of Models.** Prediction accuracy is evaluated by first calculating and then, aggregating prediction errors for the out-of-sample data using a metric, such as MSE, root mean squared error, or mean absolute deviation (Hastie et al. 2013). Estimated error is calculated as the difference between the estimated score and the actual score:

$$e = Y - \hat{Y}. \tag{1}$$

Thus, when evaluating the prediction error of outcome constructs, we would need both *estimated* construct scores and *actual* construct scores. A practical feature of composite modeling methods is that estimating model parameters yields determinable composite scores for the constructs (Steiger 1996, Rigdon 2012). These postestimation composite scores could serve as the estimated construct scores ($\hat{Y}$). However, the data themselves do not include actual construct scores ($Y$), and this poses the first challenge in bringing predictive thinking to construct-based modeling.

We propose that in the absence of actual construct scores, we can take the best possible estimate of population composite scores to evaluate the predictive qualities of constructs. Composite scores are simply weighted sums of their measurement items, with the weight parameters estimated by the composite estimation algorithm, such as PLS-PM or GSCA. These composite modeling techniques produce asymptotically consistent and unbiased estimates of the item weights (measurement model parameters) (Dijkstra 2009). Both the in-sample and out-of-sample data sets are candidates for estimating weight parameters. However, the best estimates of population weight parameters are those estimated from fitting the entire sample at hand—including all cases from both in-sample and out-of-sample data sets. This approach generates the least biased and most consistent estimates of actual composite scores and allows for consistent comparisons of prediction error across subsets of the data. We denote this estimated-actual composite score as $Y^*$, using an asterisk to denote that it is an estimated score.[1]

With this estimated-actual score, we can generate the fit and prediction errors for each outcome construct and aggregate them across all cases by using a type of MSE metric similar to those used in econometrics, statistics, and machine learning:

$$\mathrm{MSE} = \frac{\sum_{i=1}^{n} (Y_i^* - \hat{Y}_i)^2}{n}. \tag{2}$$

It is not always practical to expect separate in-sample and out-of-sample data sets to be available when evaluating a construct-based model. Thus, we first consider what to do when only one sample is available. Using a simple "holdout" validation strategy (Devroye and Wagner 1979), the cases can be randomly split into in-sample and out-of-sample data sets. The model's parameters can then be estimated on the in-sample data set, and prediction error can be computed using the out-of-sample data set. However, such single-point estimates of out-of-sample error can suffer from high levels of bias and variance (Hastie et al. 2013). Therefore, crossvalidation is often used to generate a single best estimate of out-of-sample error (Arlot and Celisse 2010, Hastie et al. 2013). Crossvalidation makes use of either random or purposive sampling and data-splitting techniques to generate multiple pairs of in-sample and out-of-sample data sets and averages the prediction error generated from these subsamples (Arlot and Celisse 2010).

Leave-one-out crossvalidation (LOOCV) iteratively produces in-sample sets of all but one case, which is used as the out-of-sample set, to provide an estimate of prediction error. Zhang and Yang (2015, p. 105) investigated crossvalidation in least squares linear regression and agree with other studies, like Burman (1989), in finding that "among the $k$-fold CVs, in estimating the prediction error, LOO (i.e., $n$-fold CV) has the smallest asymptotic bias and variance." Thus, we too suggest conducting LOOCV and reporting crossvalidated $MSE_{(out)}$ for aggregate prediction error. Nonetheless, we acknowledge that where data sets are extremely large or predictions are automated in a production environment, $k$-fold crossvalidation might be an adequate practical alternative.

**Predictive Overfit of Models.** Model overfit is evaluated by comparing model fit error with crossvalidated prediction error (Hastie et al. 2013). Despite the popularity of the $R^2$ metric, we wish to evaluate the fit of the outcome constructs using the same metric as the prediction accuracy and thus, propose the use of $MSE_{(in)}$. $MSE_{(in)}$ can be thought of as the error variance of in-sample fit (or variance unexplained) and is highly relatable to $R^2$ for standardized construct scores, such as those produced by PLS-PM, where $MSE_{(in)} = 1 - R^2$.

In order to quantify overfit, a simple exploratory step could be to look at the ratio describing the proportional change in prediction error from in-sample to out-of-sample data. This *overfit ratio* considers the difference of fit error and prediction error, relative to the fit error (Equation (3)). This ratio could be interpreted as a percentage difference in out-of-sample prediction error versus in-sample fitting, or loss of out-of-sample generalizability due to overfit:

$$\text{overfit ratio} = \frac{MSE_{(out)} - MSE_{(in)}}{MSE_{(in)}}. \tag{3}$$

This type of overfit ratio could have meta-analytic value in the long run if it is widely reported in studies across domains with different statistical and empirical properties. Thus, we encourage researchers to report both $MSE_{(in)}$ and $MSE_{(out)}$ as well as the overfit ratio in their studies. These two out-of-sample metrics can provide insight when comparing models that share the same focal outcome construct. A model with lower $MSE_{(out)}$ should be preferred in terms of out-of-sample generalizability. However, if two comparable models have similar $MSE_{(out)}$, then choosing the model with the lower overfit ratio could be justified.

However, we must caution that $MSE_{(in)}$, $MSE_{(out)}$, and any kind of overfit metric do not have any particular thresholds that can categorically demonstrate predictive value or suggest that overfit is not an issue. For example, we cannot say that an overfit ratio of 10% is sufficient or insufficient and more importantly, what that value might imply for the stability of the path coefficients or what new insights might be learned. Similarly, it might be tempting to derive a statistical test of overfit based on our earlier discussion, but such a test will yield only a meaningless categorical result that is prone to sample size considerations and $p$ hacking. Thus, we strongly discourage tests of overfit or predictive accuracy. Instead, the COA framework pursues developing a new explanatory-predictive framework aimed at extracting theoretical insight from overfit in explanatory studies.

### 3.2. Applying Predictive Methods for Theory Development

**Learning from Predictive Deviants.** The observable cases of construct-based studies (e.g., survey respondents) provide quantitative responses to measurement items. On average, researchers expect the response patterns of cases to follow the causal patterns of the hypothesized structural model. From an inferential model-fitting perspective, those cases whose responses do not follow expected patterns, such that their outcome scores are contrary to other cases with similar antecedent scores, are said to exhibit *misfit*. However, from a generalizability perspective, cases that behave substantively differently than a model specifies can also lead to an overfit situation where in trying to accommodate unusual cases, the model's ability to predict cases not in the original sample deteriorates.

We must distinguish cases that exacerbate overfit from cases that are simply misfit in sample because these two qualitatively different types of cases have different implications for out-of-sample model performance. When we consider overfit, we are most interested in a special set of cases that defy the conventional logic of fit. The cause of overfit lies in part with the overparameterization of the model, which allows for improving in-sample model fit even in the presence of unusual cases at the price of worsening out-of-sample predictive performance. Thus, our eventual goal is to identify estimated model parameters that are the result of overfit. However, we start our investigation of overfit by identifying cases that appear to fit the model reasonably well when used in-sample but whose outcomes are predicted relatively poorly out-of-sample.

We will refer to cases whose ordinary in-sample error belies their exceptional out-of-sample error as *predictive deviants*. This term draws inspiration from the notion of positive deviance (Zeitlin et al. 1990, Spreitzer and Sonenshein 2004) found in qualitative research traditions, where cases that intentionally achieve desired outcomes by exceptional means are studied to infer fruitful directions for renewed theory and practice (Pascale et al. 2010, Walls and Hoffman 2012). Our approach also follows the call of Gray and Cooper (2010), who suggest investigating unpredictable cases to test the boundaries of theories and reveal new insights. Similarly, we expect predictive deviants to inform us of interesting cases from which we could infer explanations for behaviors and outcomes that diverge from expectations.

**Predictive Deviants Are Not Outliers.** Predictive deviants represent a fundamentally new class of interesting cases compared with traditional outliers or other influential cases. The study of individual cases with incommensurate impact on a model is a well-developed stream within the construct-based modeling literature (Pek and MacCallum 2011, Aguinis et al. 2013). Such cases are thought to have *leverage* and/or *influence* or of being *outliers* (Montgomery et al. 2012). Studies have suggested best practices to not only detect but also, appropriately handle such cases (Aguinis et al. 2013, Gibbert et al. 2020). Aguinis et al. (2013) provide a comprehensive review of the definition of outliers for construct-based modeling and identify three primary types of outliers: (1) general outliers, (2) model fit outliers, and (3) prediction outliers. At first glance, the goals of this study appear to match the efforts of identifying prediction outliers. However, after contending with some conflation of terms in the literature, we will more clearly see how predictive deviants differ from the three classes of outliers.

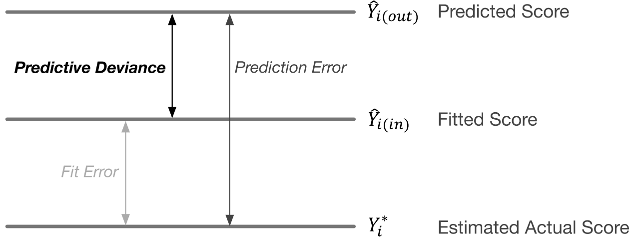
We start by looking at the terminology used to describe unusual cases, which are characterized as general outliers, model fit outliers, and prediction outliers. Outliers are generally defined as cases that are exceptional relative to other data points; they lie at a significant distance from other data points in the sample and are independent of any model (Montgomery et al. 2012). These cases are identified using such methods as leverage or Mahalanobis distance in a multivariate space or using percentiles and distance from the mean in univariate space (Montgomery et al. 2012, Aguinis et al. 2013). The framework provided by Aguinis et al. (2013) categorizes these outliers as either error outliers or interesting outliers and subsequently, provides advice for dealing with both types. Our notion of predictive deviants is easily distinguished from these general outliers because predictive deviants are very specific to the performance of a model and will differ across models even for the same data set. Moreover, predictive deviants cannot be identified by the traditional methods described for identifying general outliers because detecting predictive deviants requires model-specific methods.

In the context of construct-based models, Aguinis et al. (2013) define two categories of unusual cases that are model specific and thus, thought of as influential cases. Model fit outliers are cases that affect in-sample model fit, as measured by changes in $R^2$, $\chi^2$, root mean square of error of approximation, and so on. Prediction outliers are cases that affect in-sample parameters relating to predictors in aggregate (i.e., measured by Cook's distance or generalized Cook's distance). The predictive deviants we are interested in, for assessing out-of-sample generalizability, are neither model fit outliers nor prediction outliers because they focus on out-of-sample predictive evaluation at the case level.

The distinction between our conception of predictive deviants and the outliers considered by previous studies of construct-based modeling is visualized in Figure 3 and further discussed in Online Appendix A. Model fit outliers and prediction outliers are concerned with fit error (or the in-sample parameters involved in the estimation of fit error), which is the difference between the case's actual outcome score and its in-sample fitted score. Predictive deviants exhibit unusual predictive deviance (PD), which is the distance between prediction error and fit error, thus highlighting the change in prediction accuracy between in-sample and out-of-sample scores at the case level.

**Predictive Deviance Is Not Heterogeneity.** We also distinguish predictive deviants from unobserved heterogeneity because these two types of phenomena are conceptually and algorithmically distinct. Heterogeneity manifests as segments of cases that display behavior that is homogenous within its identified subset, whereas predictive deviants are evaluated on an individual basis and no segment is generated. We will describe later how predictive deviants can be grouped, but such groups are small and even together, constitute a small part of the data. In contrast, when heterogenous segments are all combined, they entirely encompass the data. Most importantly, unobserved heterogeneity is derived from the effect of data segments on in-sample model parameters, whereas predictive deviance is based on the difference of in-sample versus out-of-sample case-level performance of the model. In Online Appendix B, we review the conceptual and operational differences between predictive deviance and heterogeneity

**Figure 3.** Predictive Deviance




*Notes.* This quantification of influential predictive cases removes the conflation and refocuses evaluation of cases with predictive influence on the case-level, out-of-sample context. In Online Appendix A, we empirically demonstrate that predictive deviants are different from cases detected as outliers or influential cases using traditional outlier detection methods, such as Mahalanobis distance, leverage, and Cook's distance.

and empirically demonstrate that predictive deviants do not necessarily correspond to heterogenous cases.

**Defining Predictive Deviance.** We saw in Figure 3 that PD can be expressed as the difference between fit and prediction errors, and it can be written more formally as

$$PD_i = (\hat{Y}_{i\,(in)} - Y_i^*) - (\hat{Y}_{i\,(out)} - Y_i^*) \quad \forall i \in \{1, \ldots, n\}. \tag{4}$$

We can see from this formulation that PD can take positive and negative values, depending on the sign of the fit and prediction errors. Also, the $Y_i^*$ terms in Equation (4) cancel out, and we can express PD more simply as

$$PD_i = \hat{Y}_{i\,(in)} - \hat{Y}_{i\,(out)} \quad \forall i \in \{1, \ldots, n\}. \tag{5}$$

We can thus operationalize *predictive deviants* as cases with extreme absolute PD values or an extreme difference between their in-sample fitted score $\hat{Y}_{i\,(in)}$ and out-of-sample predicted score $\hat{Y}_{i\,(out)}$. This formulation is similar in spirit to the difference in fit (DFFIT) metric proposed by Belsley et al. (2005) for regression models. However, unlike regression, we cannot directly estimate a DFFIT metric for complex construct-based models and must instead devise a nonparametric approach.

**Identifying Predictive Deviants.** To identify predictive deviants, we follow three steps.

1. Conduct LOOCV to generate two scores for each case $i$ for the focal outcome constructs:
   a. the fitted construct score $\hat{Y}_{i\,(in)}$ that is estimated for case $i$ on the in-sample data set and
   b. the predicted construct score $\hat{Y}_{i\,(out)}$ for case $i$ when it is excluded from the in-sample data set but included in the out-of-sample data set.
2. Calculate a vector of predictive deviance values, which consists of the difference of the fitted and predicted construct scores for all cases in the data set (Equation (5)).
3. Identify the subset of predictive deviants with the most extreme values by taking a given percentile of the most extreme predictive deviance from both ends of the predictive deviance spectrum. We call these regions the *deviant zone*. For example, a 5% deviant zone could use symmetrical percentiles such that it consists of upper and lower regions of the form $PD_i < percentile_{2.5}$ or $PD_i > percentile_{97.5}$. Using percentiles avoids assumptions about the PD distribution.

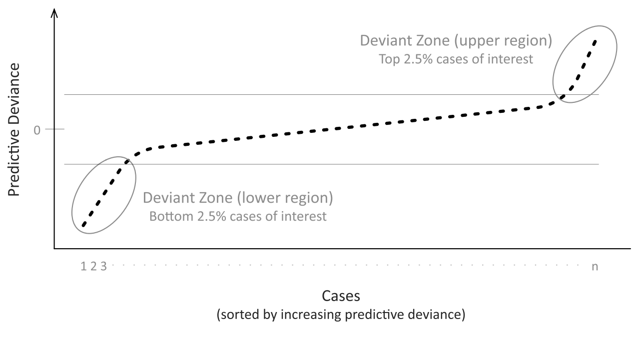
These steps identify cases where the model's ability to fit is farthest from its ability to predict. The predictive deviance of these cases implies that the model's fit to the phenomenon of theoretical interest changes in the presence or absence of these deviant cases, and so, these cases are candidates to help us find unexpected and interesting theoretical phenomena. Although we have mentioned the extreme 5% of cases as an example, we suggest that researchers visualize cases ordered by their predictive deviance to determine the suitability of the chosen percentile and the symmetry or asymmetry of the upper and lower regions of the deviant zone. For example, in Figure 4, if cases were sorted by predictive deviance and graphed against their actual predictive deviance, one should ascertain that the 5% most extreme positive or negative deviants appear to break noticeably from the middle 95% of cases in terms of their predictive deviance magnitude, loosely akin to how one might look for sharp breaks in a screeplot. This visual manner of defining the threshold for predictive deviants lets the data inform our decision making, rather than depending on ad hoc thresholds for defining the deviance zone. Our suggestion of 5% most extreme predictive deviance should thus be seen as a conservative number that researchers can adjust as needed (e.g., take 10% of extreme predictive deviance or choose a larger upper or lower region).

Figure 4 also reminds us that values of predictive deviance can be either positive or negative. *Positive predictive deviance* indicates those deviants that are underpredicted out of sample, whereas *negative predictive deviance* indicates those deviants that are overpredicted out of sample. Both situations can have ramifications that can be elucidated using domain knowledge and theory. We will demonstrate such an analysis in our later empirical example.

**Deviance Trees—Finding and Describing Deviant Groups.** The set of predictive deviants, as identified quantitatively and visually, includes those that researchers should more closely examine to determine *why* they deviate and thus, better understand the root causes of overfit. However, trying to extract theoretical implications from each deviant individually would be both time consuming and nongeneralizable. Any single deviant may be a rare occurrence that may not appear in other data sets. Instead, we strongly suggest that researchers examine natural groupings among the predictive deviants that can be described more generally.

Options abound regarding how one might group predictive deviants. However, any technique should find and describe groups of deviants based on two criteria:

**Figure 4.** Predictive Deviance of Cases (Sorted by PD Value) and Deviant Zones



(a) Each group of deviants should internally cohere in terms of their predictive deviance (closer case-wise predictive deviance within a group is better), and (b) patterns of construct scores should distinguish each deviant group from all other groups and cases (more distinct differences between groups are better). The first criterion helps us determine if the predictive deviants within a group belong together in terms of their impact on overfit and thus, converge as a group. The second criterion requires that groups of deviants are discriminable from other cases or other groups of cases. Using predictive deviance as a metric in this grouping process is essential to ensuring that our groups reflect overfit issues. Using construct scores as descriptive variables in this process is also important because constructs are the theoretical concepts of interest to researchers, and domain theory of the constructs can be applied in explaining the emergence and behavior of deviant groups.

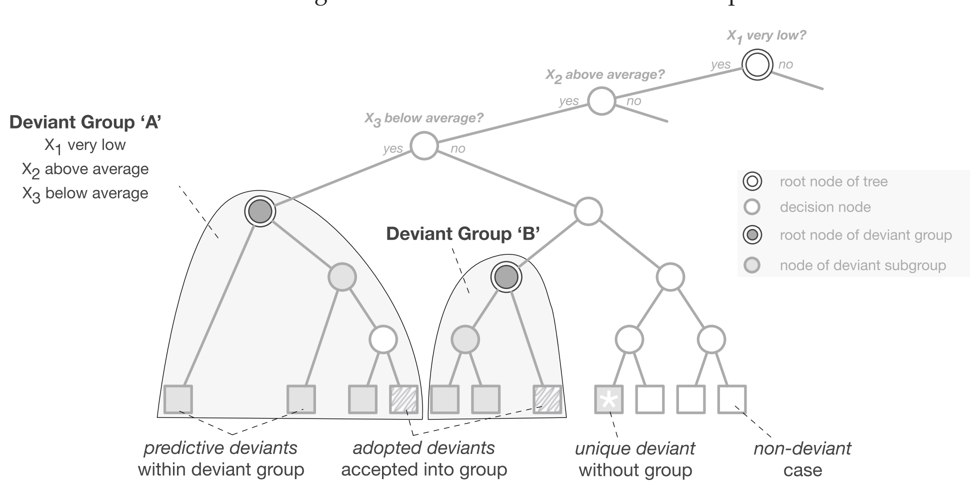
We propose here a novel technique, which we call the *deviance tree*, that is custom tailored to discover deviant groups using the two criteria. Our method is an adaptation of the general method of decision trees (and of regression trees specifically) that are used in predictive modeling where predictor variables are related to an outcome variable. Decision trees split a sample into successively smaller subsets based on decision rules, such that the final structure of splits is described by a tree where each node is a splitting rule and whose leaves are the outcome scores of cases. A major boon granted by decision trees is that the set of splitting rules from the root of a tree to a particular node tells us a descriptive story of that node. Decision trees gain generalizability by pruning decision nodes that exceed given complexity parameters, thus naturally splitting cases into distinct groups with their own rules.

Our proposed method of deviance trees is an adaptation of well-understood regression trees, but it also departs from them on critical points. Whereas implementations of regression trees traditionally model how the scores of predictor variables (***X***) can predict the scores of an outcome variable (***Y***), deviance trees model how both antecedent construct scores (***X***) and outcome construct scores (***Y***) collectively describe the predictive deviance of cases (***PD***). Thus, our method seeks to explain predictive deviance using all the theoretical constructs of a construct-based model.

Deviance trees use the same variance reduction method of identifying splitting criteria that regression trees typically employ. However, deviance trees model predictive deviances rather than outcome scores. Furthermore, whereas applications of regression trees typically prune by retaining only nodes that meet a given complexity criterion (balancing node homogeneity and tree size), deviance trees prune by retaining nodes whose underlying cases have an *average predictive deviance* that exceeds a given threshold and falls within the deviant zone. We use the same predictive deviance threshold as that used earlier to identify predictive deviant cases (e.g., largest 5% absolute predictive deviance). Our pruning logic yields deviant nodes consisting of cases that represent groups of deviants whose construct scores explain similar predictive deviance as others of their group. We note that pruning can be accomplished using stopping rules to avoid unnecessary splits or by backward pruning after the sample has been fully split into leaves with individual cases (see Witten et al. 2017 for more details on this operational distinction). We employ backward pruning that is tailored to maintaining leaves with predictive deviants. This focus on a small number of leaves makes the deviant tree more robust than regression trees. We also note that nondeviant cases often require 12 or more splits before being allocated to a leaf node. However, predictive deviants, by virtue of being anomalous (Liu et al. 2008), are quickly identified by the deviance tree in as few as three to five splits. We detail basic robustness analyses of deviance trees in Online Appendices C–F.

There are many ways of operationalizing the logic of deviance trees because there are many ways to implement

**Figure 5.** Section of a Deviance Tree Illustrating Criteria and Structure of Deviant Groups



decision trees in general. Rather than specifying one fixed process or algorithm, we formally describe how to identify cases, groups, and descriptive characteristics of interest (see Figure 5). In Online Appendix F, we consider how different implementations impact our findings.

1. For each parent node, we compute the *average predictive deviance* of all descendent cases.

2. *Deviant groups* are then subtrees rooted at nodes whose average predictive deviance lies in the deviant zone (e.g., highest and lowest 2.5% regions) and whose ancestors (nodes on the path back to the root of the tree) are not themselves roots of a deviant group.

3. For each deviant group, all its underlying cases found in the leaf nodes are deviants that belong to this group; cases in these sets that are not originally predictive deviants themselves (outside the deviant zone) are hereby termed *adopted deviants* and treated as full members of their respective deviant group.

4. The *descriptive characteristics* of a deviant group are the combined splitting decisions taken at each of its ancestor nodes. These rules describe the unique properties of a deviant group in terms of the antecedent constructs ($X$) scores. Rules that describe the same construct may be consolidated to produce a simpler list of deviant group rules. The resulting rules uniquely distinguish the underlying cases of the group from all other cases in the data set.

5. Decision nodes discovered under a deviant group (e.g., in a backward pruning operation) that also exhibit average predictive deviance in the deviant zone are *deviant subgroups*; they are subsumed by their larger deviant group and are not examined further.

6. Predictive deviant cases that are not in any group are termed *unique predictive deviants* as they do not share construct score patterns with any other predictive deviant; these unique deviants may not generalize well to other data sets and are, therefore, not examined further.

We note that some nondeviant cases are adopted into deviant groups and thus, used in further analysis, whereas some uniquely deviant cases not related to any of the groups are omitted. For an adopted deviant to be grouped with other predictive deviants, its construct scores must follow the pattern of the deviant group's rules, and its predictive deviance will often lie just outside the deviant zone. Conversely, unique deviants do not share enough common characteristics with any particular group, despite lying in the deviant zone.

Overall, it is beneficial to accept adopted deviants and reject unique deviants in further analysis as it deliberately softens the sharp rules of determining the deviant zone. Adopted deviants can help a group generalize to other data sets because it conservatively lowers the average predictive deviance of a group while accepting cases that most often lie close to the deviant zone. Rejecting unique deviants also increases generalizability by avoiding cases that are less likely to appear in other data sets. Thus, blurring the boundary of predictive deviance follows in the major spirit of this study to avoid overly strict or arbitrary thresholds and to enhance generalizability.

Altogether, a pruned deviance tree identifies deviant groups whose underlying cases share common characteristics unique to that group. Each deviant group can be quantitatively characterized by the splitting decisions that lead from the root of the tree to the root of the group. This description is data driven yet utilizes the construct scores of the theoretically important concepts of the model, making it relevant to theorists dealing with domain issues of measurement and causal relationships. Finally, we enhance the generalizability of deviant groups by selectively accepting and rejecting cases based on quantitative heuristics.

We earlier described overfit as a loss of out-of-sample generalizability that is specific to a combination of sample data and model parameters, and this can be seen from two perspectives. First, overfit can be blamed on the presence of predictively deviant cases that force the estimation process to alter model parameters to overly cater to these cases. The deviance tree procedure identifies the predictively deviant cases that exacerbate overfit. The deviance tree also groups these deviants together and gives us a set of simple, univariate descriptions for each group. These descriptions of deviant groups are important because they are the conditions under which the model begins to lose out-of-sample generalizability. We will discuss later how to derive such descriptions and demonstrate this process in Section 5.

Conversely, overfit can also be blamed on an overspecified model that is inflexible to unusual cases. The descriptions of deviant groups do not directly afford us the ability to identify which part of a model is overspecified. The deviant groups must be further exploited to isolate which of the proposed interconstruct relationships in a model are lacking out-of-sample generalizability. Therefore, the next consideration of our COA framework is to uncover which of the hypothesized relationships becomes unstable depending on the presence or absence of the deviant groups.

**Unstable Paths—Structural Relationships That Generalize Poorly.** Empirical researchers using constructs in management research are most interested in the hypothesized relationships between constructs and the generalizability of these inferred relationships. More broadly, the scientific community is increasingly aware of the importance of the predictive abilities of inferential models and calls on researchers to (a) ensure that complex path models are robust to perturbations in data and (b) seek out potentially unstable paths in causal chains (Subbaswamy and Saria 2019). However, the means of identifying unstable paths are specific to different modeling techniques. Within the COA framework for composite models, we reinvestigate the parameterized relationships in construct-based models that are perturbed by the presence of predictive deviants.

Parameters in construct-based models are either those that define the measurement model of constructs or the structural relationships between constructs. Certain structural parameters, such as interconstruct paths ($\boldsymbol{\beta}$), are directly associated with hypotheses. Identifying and examining which paths are most sensitive to the perturbations of these predictive deviants can give us insight into the boundaries of theories and potentially which hypotheses might lose out-of-sample generalizability. Researchers might then question theory, define edge cases, or seek new theories for such misbehaved parameters (Gray and Cooper 2010).

To identify unstable parameters affected by predictive deviants, we first estimate the model including all available cases. We then gather our major deviant groups identified earlier and reestimate the model successively without each of the deviant groups. We use deviant groups instead of each predictive deviant individually because the deviants within a group behave similarly in terms of predictive deviance (i.e., positive versus negative deviance) for similar reasons, and so, we avoid arriving at similar findings as we would if we examined them individually. Furthermore, removing a group at a time gives us a better idea of how the phenomenon manifesting in each group affects overfit when it is entirely removed, whereas removing only a single member of a deviant group leaves much of the group's impact on overfit intact.

Our approach is analogous to difference in beta, metrics in an ordinary regression context (Belsley et al. 2005) but differs in two key aspects; we must take a nonparametric approach because of the different nature of composite models, and we consider groups of cases to capture entire classes of reasons for overfit. We call this process *leave deviant group out* (LDGO):

1. Use the full data set $D$ to estimate the parameters of model $f(D)$, and bootstrap the data to determine coefficient magnitudes, confidence intervals, and statistical significance.
2. For each deviant group identified earlier, repeat the following.
    a. The deviant group ($g$) is excluded from $D$ to create a subset $d_{(g)}$.
    b. Reestimate the model parameters on subset $d_{(g)}$ to generate estimated model $\hat{f}(d_{(g)})$, and conduct a bootstrap to compute coefficient magnitudes, confidence intervals, and statistical significance level for model $\hat{f}(d_{(g)})$.
    c. Compare coefficient magnitudes, confidence intervals, and statistical significance levels between estimated models $\hat{f}(D)$ and $\hat{f}(d_{(g)})$.
3. The comparison in step 2(c) allows for determining which coefficients exhibit the greatest relative change in magnitude, change in level of significance, or other criteria relevant to the researcher, due to each deviant group.

The structural paths of the model that correspond to the largest coefficient changes are the unstable paths that are most at risk for performing poorly out of sample than expected from simple in-sample fitting. These paths can exhibit weak out-of-sample generalizability. Although their in-sample inferential properties are unaffected, researchers cannot draw conclusions about the predictive properties of these paths. When performing LDGO, we might find that paths with statistically significant in-sample coefficients suffer in terms of out-of-sample generalizability.

We note that it is possible for conditions such as multicollinearity and suppression effects to impact estimated model parameters. Although individual coefficients are affected by collinearity in a model, predictions are not (Makrikadis et al. 1998, Shmueli 2010). Thus, the calculation of predictive deviance and unstable paths should not be affected by issues relating to collinearity of items or constructs. However, given that collinearity can impact the variances of model coefficients, methodologists suggest diagnosing such issues first using correlation tables and metrics, such as variance inflation factor (Hair et al. 2021). We recommend investigating unstable paths only after meeting minimum criteria for both structural and measurement model evaluation (Hair et al. 2021).

**Descriptive Analysis—Investigating *When* and *Why* Overfit Impacts Theory.** Gray and Cooper (2010) suggest two essential, intuitive questions that should be asked of a theory and especially of a mature theory: "When doesn't it work?" and "why not?" They suggest that pursuing these questions is the best method to discover auxiliary hypotheses and boundary conditions. Researchers should use the limitations, where the data do not conform to the theory, to test the boundaries that indicate where a theory would or would not hold (McGuire 1983, 1989). The most basic approach to examining these boundary conditions and auxiliary hypotheses is to carefully scrutinize those situations that are poorly predicted (Gray and Cooper 2010).

In this spirit, researchers can derive a *descriptive analysis* of when, how, and why overfit implies lowered out-of-sample generalizability by examining each deviant group's descriptive characteristics and corresponding unstable paths. Researchers can start by using the distinguishing rules of each deviant group, which are expressed in terms of construct scores, to quantitatively characterize each group and so, state *when* overfit occurs. The collection of conditional statements at each decision node, from the root of the deviance tree to a particular deviant group, describes the group of deviant cases based on construct scores. These univariate rules are easy for domain experts to understand and interpret because they relate to theorized constructs and are expressed relative to standardized scores. Thus, researchers can express the precise conditions (e.g., "high levels of behavioral intention and low levels of social influence") under which overfit manifests and the theorized model begins to lose out-of-sample generalizability. The unstable paths corresponding to each deviant group in LDGO can tell us *how* overfit affects the model at hand. Here, we suggest identifying borderline significant paths that might lose significance in the absence/presence of a group, which should be cautiously interpreted. Researchers can also identify strong, significant paths that exhibit the greatest instability, which might suggest important boundary conditions or new phenomena. Using domain knowledge, researchers can then infer *why* these deviations happen and whether to rethink theory. We further encourage researchers to collect open-ended or structured qualitative data in their surveys that allow for a supplementary qualitative investigation to reinforce descriptive explanations. Importantly, we strongly caution researchers against eliminating predictive deviants, individually or as a group, from their data set without fully understanding the reasons for their impact on the model.

Following the best practice of examining the robustness of machine learning methods under different algorithmic choices (D'Amour et al. 2020), we explore the results of the COA framework when applied under varying conditions and algorithms in Online Appendices C–F. We find the framework robust for the estimation of prediction metrics, identification of predictive deviants, deviant groups, and group rules across composite modeling algorithms (PLS-PM and GSCA) (Online Appendix C) and deviance tree algorithms (Online Appendix F). Further, we find that the framework is stable when both the nondeviant data and structural model are slightly perturbed (Online Appendices D and E).

In summary, we propose an explanatory-predictive framework that allows researchers to evaluate and identify sources of overfit in construct-based models and infer implications to theory. The COA framework identifies groups of deviant cases that exacerbate overfit and the structural relationships between constructs that may not generalize well out of sample. These results can guide a descriptive analysis exploring the deeper reasons for such conflicts. Researchers can employ this framework to leverage predictive methods for theory development.

# 4. Empirical Demonstration—Project Management Software Adoption

To showcase the COA framework and demonstrate how it might lead to theory development and discovery, we apply the method in an empirical example. We will show how researchers applying construct-based path modeling should take advantage of the predictive qualities of their work. This demonstration can serve as a template for researchers to report results and implications of predictive deviants and unstable paths. We conduct the quantitative portion of the analysis using programmatic routines we have created for the R statistical environment (R Core Team 2022) to implement the COA framework.[2] Our routines use the R package SEMinR (Ray et al. 2022) that provides parameter estimates of PLS-PM models. By estimating parameters, we produce fitted models that also serve as trained models for prediction.

## 4.1. Context and Data

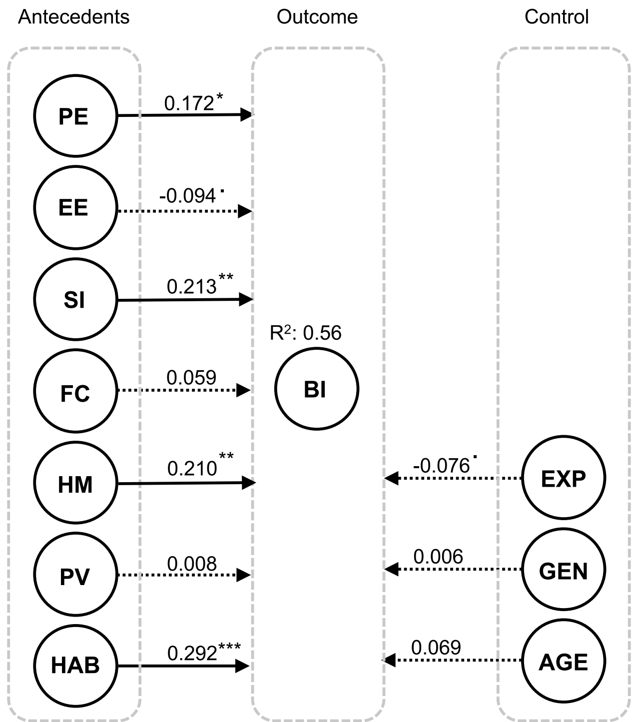
We reevaluated a simplified but major portion of the well-regarded UTAUT2 model (Venkatesh et al. 2012) that seeks to explain information system adoption and use. To enhance the possibility of novel findings, we chose an interesting but lesser studied context in which to reexamine the theory: personal task management software. We adjusted some measurement items from their original survey instruments to match this context. We asked respondents to consider the Trello (https://trello.com/) software as the focal information system. Trello allows users to manage projects using a card-based Kanban format (Junior and Godinho Filho 2010), popular in software project management methodologies, detailing the tasks to be completed along with member assignment and completion dates. Trello also allows users to collaboratively manage and monitor personal tasks and projects—a context where the use of such software is voluntary. We sought to ascertain which factors were associated with respondents' intentions to adopt and use Trello in the near future. Additionally, we were especially interested in how the antecedent constructs of UTAUT2 generalize out of sample in predicting behavioral intentions. Respondents were shown an introductory tutorial video on using the software produced by Trello and then presented with our survey instrument, which contained measures of constructs seeking to explain respondents' future intentions to adopt and use Trello.

Figure 6 illustrates the conceptual model of the empirical demonstration. The primary antecedent constructs are performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), hedonic motivation (HM), price value (PV), and habit (HAB), which collectively explain the behavior intention (BI) to use Trello. We control for variables such as experience, gender, and age. Although the UTAUT2 model includes the control variables in moderating roles, we have excluded these relationships to keep our demonstration simple.

## 4.2. Results and COA

We collected data for this demonstration context using an online survey and recruited participants from the

**Figure 6.** Estimated Model for Project Management Software Adoption




*Notes.* Path significances are indicated. EXP, experience; GEN, gender. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; •$p < 0.10$.

Amazon Mechanical Turk (AMT) platform. We targeted North American respondents and offered a small monetary reward. We ran an initial pilot test of 50 survey responses with open-ended questions assessing the clarity of our instructions, measurement items, and other potential issues in the survey design. We sent out a final survey instrument to 250 AMT respondents who qualified as "masters" with high performance in prior studies, who were located in the United States, and who did not participate in our pilot test. We collected all responses within a 24-hour time frame. We rejected responses completed in an infeasibly short amount of time and those with suspicious answer patterns, leaving us with a final sample of 216 valid responses.

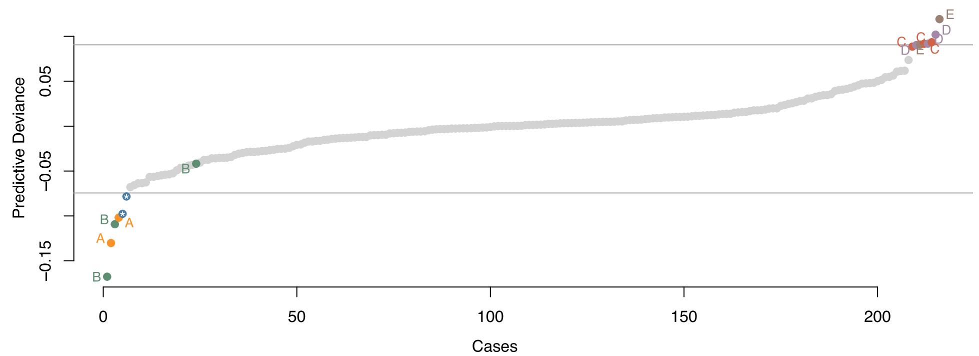
We employed PLS-PM for our composite model analysis, although other composite modeling approaches, such as GSCA or RGCCA, could also be used. Figure 6 also shows the estimated structural coefficients of our fitted model. The estimated model meets the criteria necessary to validate a composite model (Henseler et al. 2016, Hair et al. 2020). We will not discuss these validations in detail but will instead delve into the results from applying the COA framework. The parameter estimates are in accord with Venkatesh et al. (2012), with HAB, HM, SI, and PE being the principal determinants of BI.

**Prediction Accuracy and Overfit.** We calculated fit $MSE_{(in)}$ at 0.435 and performed LOOCV to obtain the prediction $MSE_{(out)}$ of 0.480. The difference of out-of-sample and in-sample MSEs was 0.045, which yields an overfit ratio of 10.3%. These figures are not immediately meaningful in our study, but we hope that as further studies report these figures, we can get a meta-analytic sense of the range of prediction error common with the constructs and given context.

**Predictive Deviance and Deviant Groups.** We applied the COA framework to analyze how and why our model might be overfit to our sample. COA begins by computing the predictive deviance of each case using LOOCV. The distribution of predictive deviance is shown in the predictive deviance plot in Figure 7. Based on the natural breaks in the predictive deviance distribution, we chose a symmetric deviant zone of the 5% most extreme cases, giving us 2.5% of cases on either end of the predictive deviance continuum.

Next, we grew a deviance tree to identify groups of similar predictive deviants. Figure 7 also shows the 13 predictive deviants belonging to five groups (A–E), including a few adopted deviants that emerged from outside our specified deviant zone but close to our thresholds. Groups A and B displayed negative deviance, and groups C–E displayed positive deviance. Predictive deviants outside these groups were not considered in our analysis as they are unlikely to generalize to other samples. The splitting criteria by which the five deviant groups emerge are shown in the structure of decision nodes of the deviance tree in Figure 8.

**Figure 7.** (Color online) Predictive Deviance of All Cases (Sorted in Increasing Values)




*Notes*. Horizontal lines indicate the upper and lower thresholds of the deviant zone. Gray points are nondeviant cases, and colored points are predictive deviants with group labels (A–E). Predictive deviants within the 95% interval are adopted deviants. Points labeled with an asterisk are unique deviants that do not belong to any deviant group.

Table 1 summarizes the descriptive characteristics of each group. Each group displays group-level metrics, descriptive characteristics, and unusual patterns that are contrary to theory.

Whereas our hypothesized model suggests largely positive, symmetric relationships (e.g., high/low SI and high/low BI), each deviant group has cases whose antecedent scores are not commensurate with their BI outcomes (e.g., low SI and high BI). Figure 9 illustrates the varying levels of construct scores (on average) for each group. High and low scores can be observed above and below the gray bands indicating the middle 50% and 75% regions of standardized construct scores. The most unusual combination of scores for each group is described in the last column of Table 1.

**Unstable Paths.** The groups of predictive deviants influence the fitted model such that removing them causes instability in path estimates, which can affect significance and alter their impact. We performed LDGO analysis on our model and examined which deviant groups had large impacts on path coefficients (Table 2). We focus on paths that were originally significant and sustain changes of ±0.01 or higher when a deviant group is removed.

### 4.3. Descriptive Analysis of Overfit

Looking at the LDGO results path-wise, the structural paths to BI from all its antecedent variables appear to be unstable, although some are more concerning than others. PE, EE, SI, HM, and HAB were all hypothesized and significant paths that exhibit instability under various conditions. The path from EE to BI is borderline significant, and even the removal of a single deviant group (e.g., group C's three cases) could render it entirely nonsignificant; we suggest treating its statistical significance with caution and not overly stating its influence in this context. The paths from SI and HAB to BI encountered the largest path instabilities, with a single deviant group's

**Figure 8.** Conceptual Illustration of a Deviance Tree

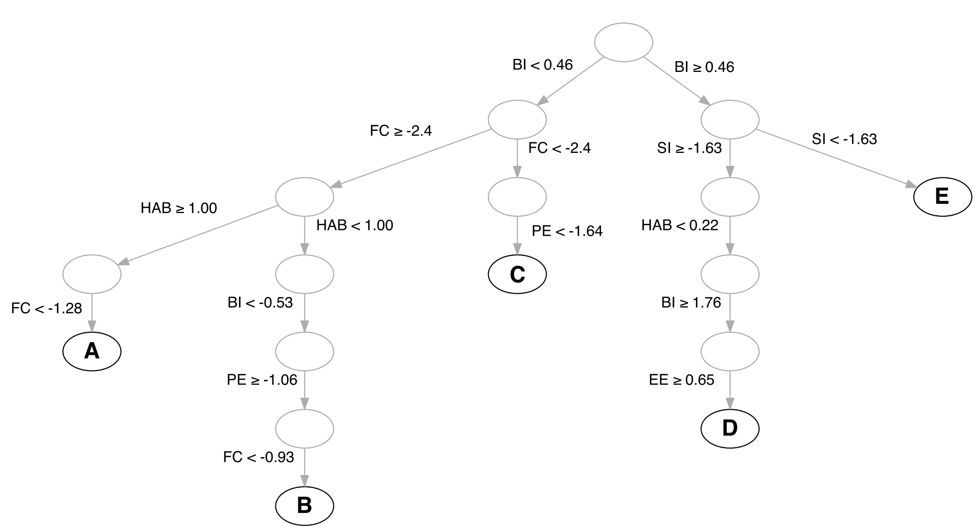
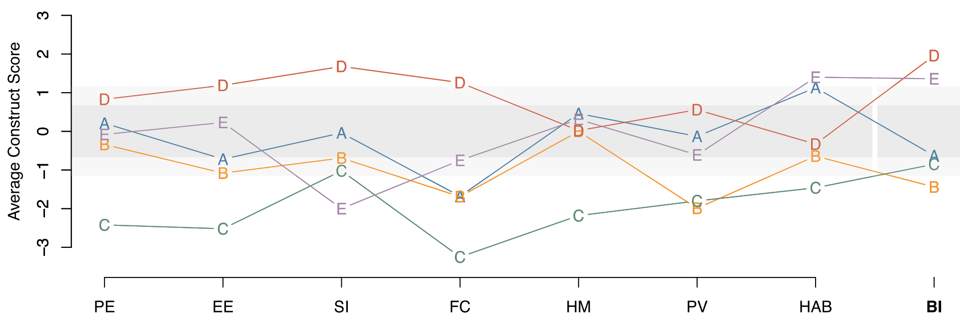


**Figure 9.** (Color online) Average Construct Scores of Deviant Groups




*Notes.* Group names (A–E) are labeled at each point. The dark gray band shows the expected middle 50% region for standardized construct scores; the light gray band shows the expected 75% region for standardized construct scores. The outcome construct is BI. Antecedent constructs are PE, EE, SI, FC, HM, PV, and HAB.

absence or presence changing the magnitude of their effect by 0.04–0.05; we need to consider the boundary conditions reflected in groups D and E, under which these otherwise strong effects appear to lose or gain effectiveness.

Taking a closer look at groups D and E, we find that each group produces instability, but group D produces widespread instability to the entire model. Compared with our earlier descriptive analysis (Table 1), users in group D exhibit below average HAB yet very high BI, which runs counter to theory and likely reduces the estimated effect of HAB on BI. When group D is left out of the sample, the effect of HAB on BI increases (Table 2). This group of potential users is willing to use Trello despite not foreseeing any need to habitually use such project management software. This finding might suggest that tools such as Trello have practical value for some users looking to use it for one-off projects or events rather than regular tasks, and information technology services should consider ways of licensing tools for such nonperpetual use. We also note that the absence of group D reduces the effect of SI on BI, suggesting that these users are among those who see great social value in these tools.

Group E exhibits very low SI and yet, high BI, which also runs contrary to theory, and its effect on the model is the opposite of group D. The presence of group E in the sample reduces the effect of SI on BI, whereas its absence increases the effect. Cases in this group appear to discount the social value of collaborative project management tools like Trello yet still intend to use them, presumably for their own individual purposes. Such users might have novel ways of deriving value from such tools, and their usage patterns deserve greater scrutiny in future. Moreover, we should not assume that social influence is a necessary condition for users to adopt this tool. From a practical viewpoint, information technology services could benefit by highlighting both types of usefulness in marketing and training—explicitly demonstrating both personal and group utility could help change this perception in some consumers and promote the overall utility. We also note that group E's absence from the sample reduces the effect of HAB on BI, which suggests that these users are among those who are most likely to use such project management tools regularly.

## 5. Discussion

The COA framework helps researchers gain theoretical insight from studying the out-of-sample generalizability of their models. Whereas inferential studies seek to test theoretical models on the average level, prediction is primarily focused on maximizing out-of-sample predictive performance at the case level, often with minimal concern for theory. We bridge these two worlds by providing an analytic framework that allows researchers to extract information of potentially theoretical value from predictive performance of models. From these predictive cues, we demonstrated how researchers can more readily expand theory and find the boundaries of their theoretical assumptions. We contribute to the predictive methodology for construct-based models by identifying and hurdling conceptual and methodological issues

**Table 1.** Description of Deviant Groups

| Group | Size | Type of deviance | Descriptive characteristics | Unusual patterns |
|---|---|---|---|---|
| **A** | 2 | Negative | **BI** $< 0.46$ and **FC** $\in [-2.45, -1.28]$ and **HAB** $\geq 1$ | Low **FC** and avg **BI** |
| **B** | 3 | Negative | **BI** $< -0.53$ and **FC** $\in [-2.45, -0.093]$ and **HAB** $< 1.00$ and **PE** $\geq -1.06$ | Avg **PE** and low **BI** |
| **C** | 3 | Positive | **BI** $< 0.46$ and **FC** $< -2.45$ and **PE** $< -1.64$ | Low **PE** and avg **BI** |
| **D** | 3 | Positive | **BI** $\geq 1.76$ and **SI** $\geq -1.63$ and **HAB** $< 0.22$ and **EE** $\geq 0.65$ | Avg **HAB** and high **BI** |
| **E** | 2 | Positive | **BI** $\geq 0.46$ and **SI** $< -1.63$ | Low **SI** and high **BI** |

*Notes.* Avg, average. The outcome construct is BI. Antecedent constructs are PE, EE, SI, FC, HM, PV, and HAB.

**Table 2.** Structural Model Coefficient Instability for LDGO Analysis

| | | Group A | | Group B | | Group C | | Group D | | Group E | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DV: BI | Original path | Δ | % | Δ | % | Δ | % | Δ | % | Δ | % |
| **PE** | 0.172* | 0.009 | 5.3 | **0.018** | **10.7** | −0.001 | −0.7 | **−0.016** | **−9.5** | 0.004 | 2.3 |
| **EE** | −0.094• | **0.015** | **−15.6** | 0.005 | −5.7 | **0.019** | **−20.6** | **−0.016** | **17.2** | −0.009 | 9.0 |
| **SI** | 0.213** | −0.004 | −2.1 | −0.005 | −2.3 | **−0.014** | **−6.5** | **−0.041** | **−19.1** | **0.046** | **21.4** |
| FC | 0.059** | −0.037 | −61.6 | −0.033 | −55.2 | 0.023 | 39.1 | −0.013 | −21.3 | 0.008 | 13.7 |
| **HM** | 0.210** | 0.007 | 3.3 | **0.023** | **10.8** | 0.003 | 1.3 | **0.026** | **12.4** | 0.006 | 2.9 |
| PV | 0.008 | −0.002 | −19.5 | −0.033 | −416.8 | 0.01 | 121.9 | −0.011 | −135.2 | 0.009 | 119 |
| **HAB** | 0.292*** | **0.011** | **3.8** | −0.005 | −1.7 | −0.003 | −1.2 | **0.047** | **16.0** | **−0.052** | **−17.8** |

*Notes*. Original structural path estimates are shown with significances. Changes in paths when each group was left out are expressed as differences (Δ) and percentage differences (%). Bold percentages indicate originally significant paths with large instability (±0.01 or more). DV, dependent variable. The outcome construct is BI. Antecedent constructs are PE, EE, SI, FC, HM, PV, and HAB.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; •$p < 0.10$.

that have thus far precluded jointly using predictive and inferential methods on construct-based models. Throughout this study, we have recommended what to do, and cautioned against what not to do, when conducting composite overfit analysis of a construct-based model. We summarize our cautions in Table 3 and our recommendations in Table 4.

### 5.1. Contributions to Methodology

Constructs are the focal variables of interest to theory researchers. Whereas studies of predictive methods in construct-based research have focused on predicting measurement items, we argue for evaluating how well we can predict the constructs that theorists are primarily interested in. In computing prediction error, we overcome the lack of actual construct scores in data by using estimated-actual composite scores from the full data sample and theorized model.

We contribute a first clear framework that outlines which aspects of predictive performance can be meaningfully and practically evaluated to benefit theory-driven, explanatory studies employing construct-based models. We caution against statistical testing of a model's predictive performance and validity and also, against the typical remedies found in predictive studies, such as dropping or regularizing parameters. These prediction-enhancing methods are contrary to the goals and aims of theoretical causal modeling.

We suggest researchers focus their investigation on the idiosyncrasies of the data and model that are producing overfit. We emphasize overfit because it describes the dangers of overreliance on fitting procedures and practices that render a model ungeneralizable to new and future data. As we demonstrated, the removal of even a single deviant group consisting of two to three cases can alarmingly reverse the significance of hypothesized paths. However, rather than testing for overfit or modifying the model to reduce overfit, we suggest analyzing the root causes of overfit by detecting and describing deviant groups and examining the unstable relationships that do not generalize well in practical, out-of-sample ways. Our direction and procedures contribute an overarching framework to develop theoretical and domain insights from these deviations and instabilities.

We developed a method to identify unusual cases that contribute to overfit. We proposed that we must identify cases that are far harder to predict out of sample than they are to fit when they are within sample. To operationalize this detection scheme, we proposed a case-level predictive penalty metric based on the difference between in-sample fitted scores and out-of-sample predicted scores for cases with good or moderate in-sample fit. Our approach gives researchers a simple and understandable procedure to highlight cases of interest in their models.

Our COA framework introduces a novel approach to identifying and describing deviant groups. Our proposed method of deviance trees offers a new descriptive

**Table 3.** Cautions When Applying the COA Framework

| What not to do | Why not |
|---|---|
| Do NOT modify your model based solely upon predictive considerations | The primary purpose of inferential modeling is theory development and explanation |
| Do NOT simply eliminate predictive deviants from your sample | Eliminating predictive deviants might appear to improve model fit and might even give the illusion of reducing overfit (e.g., overfit ratio), but doing so actually worsens overfit, deteriorates prediction of new and future cases, and hides important theoretical opportunities |
| Do NOT test or make categorical statements about predictive performance or overfit based on arbitrary thresholds | Predictive performance and overfit are not categorical; the degree of acceptable overfit (e.g., overfit ratio) is dependent on domain and what is being measured |

**Table 4.** Recommendations for Applying the COA Framework

| What to do | Why |
|---|---|
| Evaluate and report the predictive accuracy ($MSE_{(out)}$) and overfit ratio | All composite models are likely overfit to various degrees and will certainly have prediction errors. Researchers conducting model comparisons may then consult the reported metrics in extant literature to give domain- or context-specific guidelines for expected predictive performance. |
| Identify deviant groups | Groups of predictive deviants inform us of unusual behaviors and phenomena that go against the grain of the theory; they largely fit the model but would be predicted poorly. |
| Identify unstable paths | Certain paths in a model may fluctuate considerably if estimated on new data. These paths should be treated cautiously when considering practical applications of the model. If unstable structural paths relate to critical hypotheses, this should be reported in the limitations of the research. |
| Perform descriptive analysis of overfit | We need to understand what drives these deviants and why they do not conform to theory. Descriptive criteria can provide explanations for the presence of deviant groups and unstable paths. |

technique that utilizes the case-level, prediction-oriented nature of regression trees to help identify groups of cases that exhibit similar predictive deviance for similar reasons. This data-driven approach gives researchers quantifiable rules that describe groups using the scores of theoretical constructs and does so without resorting to arbitrary and inflexible thresholds. Deviance trees offer considerable promise in bringing explanatory power to studying overfit, and this paper has touched upon one possible direction.

We also proposed the LDGO procedure for identifying unstable paths—coefficients affected by overfit that are unstable in the presence of the deviant groups we identified earlier. This approach allows us to identify model relationships that may not generalize well to new or future cases. Together with the descriptions of deviant groups from deviance trees, researchers can now begin to unbundle the reasons why overfit occurs and what it implies for theory.

### 5.2. Contributions to Theory Building

From the methodological contributions, we hope to build new practices and procedures to enhance theory building. In particular, we address recent calls for bringing predictive methods to theory development in the composite modeling community (Hair et al. 2017b), and we do so with a firm focus on furthering the primary goal of empirical researchers: to build and discover theory. At the same time, we also pay heed to urgent calls in the broader scientific community to break away from categorical claims and to instead delve into the uncertainties and complexities of domain and context (Amrhein et al. 2019). We do so by bringing a mix of out-of-sample logic from predictive methodologies (Shmueli and Koppius 2011) and case-based analysis of deviance from theory (Gray and Cooper 2010). By applying inductive methods, prevalent in predictive and machine learning approaches, to deductive research, the generalizability and reproducibility of research can be improved.

In order to use case-level predictive metrics for developing theory, we proposed identifying predictive deviants and unstable paths and then, qualitatively understanding the most extreme deviant groups of cases to find possible reasons for the deviant behavior. We hope this framework provides researchers a simple and understandable procedure to target and explain groups of interest, the paths that are unstable in their presence, as well as highlight boundary cases and conditions that researchers might not have considered or anticipated. We further strongly recommend that researchers include substantial qualitative questions in surveys to allow them to deduce the nature of the deviant behavior and theorize on the instability of these model paths and the implications thereof.

We also briefly touched upon the possibility of model comparison using some of the metrics generated in the COA process. We noted that the predictive metrics are outcome specific and should only be applied to the same focal outcome construct of two or more models. The out-of-sample predictive error (e.g., $MSE_{(out)}$) serves as a direct measure of out-of-sample generalizability that can be used to compare two models. The overfit ratio serves as a measure of the trade-off between predictive accuracy and in-sample fit, and it too can be compared across models with the same outcome. However, these predictive metrics must be seen relative to relevant in-sample performance metrics (e.g., $R^2$). If two models perform similarly in-sample, then the comparison of predictive metrics is straightforward; the model with better predictive performance is superior. However, if the two models have different in-sample characteristics, then the out-of-sample performance is harder to assess; researchers would need to carefully disentangle the trade-off between fit and out-of-sample generalizability.

### 5.3. Managerial Implications of the COA Framework

Although the COA framework seeks to correct and enhance the inferential approaches of academic researchers, it can also benefit industry practice in certain scenarios. Both perceptual constructs, such as behavioral intention, and business artifact composites, such as business performance metrics, have more practical value than the items that measure them (Henseler 2017). Consequently, firms could aggregate metrics and measurements into quantifiable composite constructs and examine their relationship with other constructs.

The COA framework provides solutions for predicting and evaluating focal business concepts and allows for practical decisions and implications to be derived. We earlier highlighted how the set of rules that describe each deviant group in the deviance tree can help researchers identify boundary conditions of a theory. Similarly, if a working model of a firm's practices is defined, then the descriptions of deviant groups can suggest where a firm's assumptions break down. For example, consider a firm that has modeled how much value its clients derive (e.g., satisfaction, purchases, or repatronage) from certain policies or practices (e.g., customer support or promotions) and has estimated this model from survey data or actual figures of sales and interactions. Applying COA to the model results might identify deviant groups of clients who do not derive the benefits despite availing the policies. Moreover, certain relationships between policies and expected benefits might be unstable. Such findings might warn of unstudied circumstances where a firm's mental model of its value proposition might not apply or might highlight a need for marketing efforts to alleviate the concern of particular deviants (such as the concerns raised in the empirical example).

Alternatively, predictive deviants might sometimes exhibit desirable behaviors but for unexpected reasons and motivations—much like the positive deviants found to thrive in dire circumstances in human development studies (Zeitlin et al. 1990). In a service context, managers might identify deviant groups of customers who are eager patrons despite experiencing dissatisfaction or failure in service quality. Such customers might find other intangible or otherwise unconsidered benefits that could be applied to other cases. Managers and frontline employees might intervene in these unusual scenarios to further understand these cases and further disseminate the hidden benefits found by these customers.

### 5.4. Limitations and Future Research

There are several avenues of research worth mentioning that are beyond the scope of this study. First, we formulate the COA framework for composite models because the literature on composite modeling has clearly illustrated the potential for overfit. Yet, we do not preclude the application to common factor models, such as CB-SEM, but note that it would require several considerations beyond the scope of this article—not the least of which is the indeterminacy of the factor scores (Rigdon 2012). Only once future research develops methods of producing consistent and determinable scores for common factor models can the COA framework be applied to such models.

We have mentioned that the COA framework produces metrics of out-of-sample performance and overfit that could be used to conduct explanatory-predictive model selection, an area of methodological development that is seeing recent interest (Sharma et al. 2021). However, many nuances must be worked out to develop a full procedure for conducting such explanatory-predictive model comparisons. In particular, we mentioned that such analysis should be done on the same focal construct across models, but more research is needed to understand whether the antecedent structure of models must also be similar. Moreover, we mentioned that it is complicated to compare models with both different in-sample and out-of-sample characteristics. We believe there is ample room to develop fuller procedures that guide researchers through comparing the trade-offs entailed in such a comparison. We expect that $MSE_{(out)}$ and the overfit ratio must both play a role in this process or be the basis for more intricate metrics in this direction.

We designed the COA framework to address out-of-sample generalizability of construct-based models, but it can also be adapted to analyze any type of explanatory quantitative issue. One extension of COA that is particularly worth highlighting is adapting it to analyze how a model changes when applied to data from a different context or different time period.[3] There has been a growing interest in machine learning research on ways for tackling such *data set shift* (Quionero-Candela et al. 2009). We surmise that COA can be used to address such questions by altering its focus from comparing in-sample versus out-of-sample performance to comparing out-of-sample versus out-of-context (or out-of-time) performance. Much of what we have described in terms of identifying predictive deviants, describing deviant groups, and locating unstable paths would apply to helping to understand how a model changes with context and over time. We would start by refitting the model to the out-of-context sample. Using the out-of-context model, we would generate predicted construct scores for the original sample. This results in an out-of-context construct score $\hat{Y}_{i(ooc)}$ for each record $i$, in addition to $\hat{Y}_{i(in)}$ and $\hat{Y}_{i(out)}$ described earlier. The difference $(\hat{Y}_{i(ooc)} - \hat{Y}_{i(out)})$ captures the predictive deviance because of the new context. We can compare these deviances with the predictive deviance values: for example, by computing a *context deviance ratio* $(MSE_{ooc} - MSE_{(out)})/MSE_{(out)}$. If the original model generalizes to the new context, we would expect the *MSE* differences to be negligible and the ratio to be small. Comparing the context deviance ratio with the overfit ratio, discussed in our paper, can provide further indication of the out-of-context effect.

However, just as we cautioned when introducing the overfit ratio, such metrics for out-of-context analysis would not have any particular threshold that can categorically demonstrate predictive value. Hence, the next step is to identify context-driven deviants and groups using the same deviance tree procedure described earlier, except that we would now operate on the out-of-context predictive deviants $(\hat{Y}_{i(ooc)} - \hat{Y}_{i(out)})$. The underlying assumption is that if the model generalizes to the new context, the deviance tree will identify deviant groups with the same characteristics as before. Finding deviant groups with very different characteristics would indicate a generalization issue, which can be further investigated to identify the relevant weak paths. Thus, whereas COA currently seeks to understand overfit, the surmised comparison described here would instead seek to understand data set shift. There are of course many extra considerations that might apply to this new direction, which is beyond the scope of this study but ripe for future investigation.

Finally, the COA framework is a quantitative procedure to deriving insight from overfit, and there are other choices one could make for each quantification. However, we also hope that researchers will explore more qualitative approaches to studying predictive deviance, where the stories of individual deviants and deviant groups are more thoroughly uncovered using interviews or observation.

### 5.5. Conclusion

Predictive methods have great potential to complement inferential construct-based modeling and directly address the focal concepts that empirical researchers care most about: constructs and their hypothesized relationships. We have created an explanatory-predictive framework that provides a novel analytic tool for researchers that addresses wider concerns in the scientific community about the limitations of extant inferential methods. We believe that identifying the root causes for predictive deviance and unstable paths can be applied to any regression analysis and not necessarily limited to construct-based modeling. However, given what we now know about the prevalence of overfit in construct-based models, we feel it is imperative for this modeling tradition to pay special attention to overfit and equip itself with modern solutions to mitigate it. Armed with the new COA framework, we hope researchers can explore the boundaries of their theories and draw out more nuanced theoretical and practical implications.

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## Endnotes

[1] Estimated score $Y^*$ is produced as a function of item scores and measurement weights ($Y^* = x\hat{w}$). We are not using the typical "hat notation" of estimations for $Y^*$ to avoid confusion with fitted scores $\hat{Y}_{(in)}$ and predictions $\hat{Y}_{(out)}$, which are generated as a function of antecedent construct scores ($\hat{Y} = X\hat{\beta}$).

[2] Code and data relevant to this study are available through *Management Science* and on GitHub. The general code for the COA framework is packaged as an R library at https://github.com/sem-in-r/semcoa. It can be used on any composite structural equation model and data set. The code and data for simulations and empirical demonstrations in this paper and appendices are available at https://github.com/sem-in-r/coa_framework_paper.

[3] We thank an anonymous reviewer for bringing this potential extension to our attention.

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